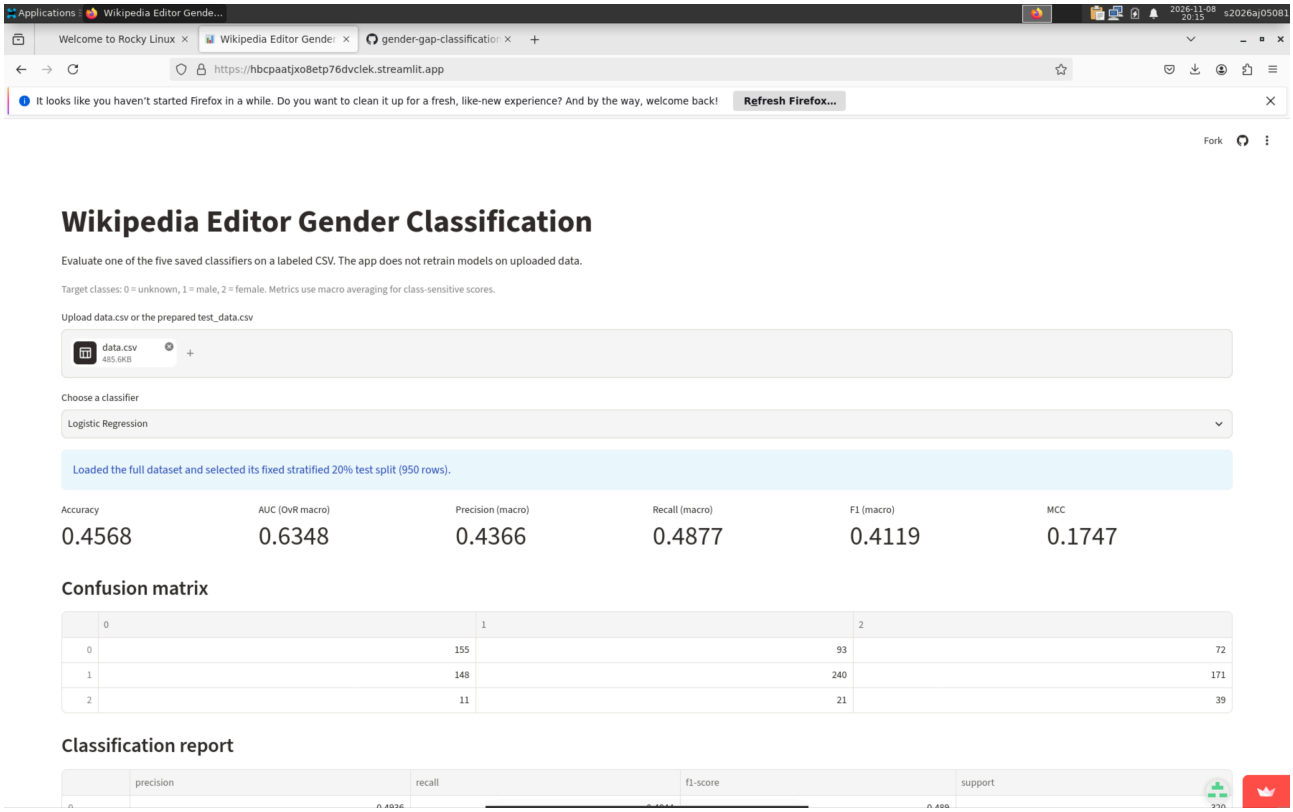
Machine Learning - Assignment 2

Project: Wikipedia Editor Gender Classification
GitHub Repository
https://github.com/amoldeokate-07/gender-gap-classification-app
Live Streamlit App
https://hbcpaatjxo8etp76dvclek.streamlit.app/

Bits Lab Portal Screen Shot



README Content

Problem statement

Compare five classification algorithms on the Gender Gap in Spanish Wikipedia dataset and provide an interactive app
for evaluating saved models on labeled CSV data. The app uses saved models and does not retrain on uploaded files.

Dataset description

Gender Gap in Spanish WP, from the UCI Machine Learning Repository (DOI 10.24432/C5XP52). It contains 4,746 records and 18 candidate input features. The target is gender: 0 = unknown, 1 = male, 2 = female. C_api and C_man are excluded to prevent target leakage. firstDay and lastDay are converted to year, month, day, and hour features. No missing values or duplicate rows were found. The split is stratified with a fixed random seed.

GitHub Repository Link

https://github.com/amoldeokate-07/gender-gap-classification-app

Models and evaluation results

AUC is multiclass one-vs-rest macro AUC. Precision, recall, and F1 use macro averaging.

Observations

Decision Tree has the highest macro F1 (0.4122). KNN has the highest accuracy (0.5537), but lower macro F1. Gaussian Naive Bayes has the highest AUC (0.6519) and precision (0.5678), but the lowest accuracy and F1. Logistic Regression has the highest MCC (0.1747). The held-out test set has 320, 559, and 71 rows for classes 0, 1, and 2. By macro F1, Decision Tree is the winner in this run.

Model	Performance observation	
Logistic Regression	Highest MCC and second-highest macro F1.	
Decision Tree	Highest macro F1.	
K-Nearest Neighbors	Highest accuracy; lower macro recall and F1.	
Gaussian Naive Bayes	Highest AUC and precision; lowest accuracy and F1.	
Support Vector Machine	Competitive AUC, recall, and F1.	
Overall winner	Decision Tree by macro F1.	

File	Purpose
app.py	Streamlit interface.
train_models.py	Preprocessing, split, training, and metric generation.
model/	Saved versions of all five classifiers.
data.csv and test_data.csv	Full source dataset and labeled held-out data.
requirements.txt and README.md	Dependencies and project documentation.

App and project details

Streamlit app link

https://hbcpaatjxo8etp76dvclek.streamlit.app/

Upload data.csv or test_data.csv, select a saved classifier, and view accuracy, AUC, precision, recall, F1, MCC, a confusion matrix, and a classification report. For data.csv, the app selects the same fixed stratified 20% test split. It does not retrain models on uploaded data.

Run locally

```
pip install -r requirements.txt
python train_models.py
streamlit run app.py
```

Repository contents

Training uses an 80/20 stratified split and random seed 42. All five models use the same split. C_api and C_man are excluded from the input features.